CASM: A CONTEXT-AWARE SEMI-MARKOV ALTERNATIVE TO DBN POST-PROCESSOR FOR BEAT TRACKING

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ABSTRACT

This paper

The last stage of beat tracking must turn dense neural activations into a discrete and musically coherent event sequence. Direct peak picking preserves local evidence but can fragment the pulse, whereas dynamic Bayesian networks (DBNs) restore continuity through globally configured tempo and meter assumptions that need not transfer to difficult music. We introduce CASM, a plug-in, context-aware semi-Markov decoder over a sparse graph of activation-supported beat candidates. CASM estimates local periodicity and its ambiguity from the current activation sequence; this ambiguity jointly controls the strength and width of each event-to-event duration potential. It therefore commits strongly when the pulse is clear and relaxes toward the neural evidence when competing metrical levels are plausible. A beat-count safeguard and a beat-synchronous meter decoder make this behaviour deterministic and auditable. One globally calibrated CASM configuration is applied without backbone retraining or track-specific BPM and meter settings to Beat This, MSCNN-lite, and TCN activations on GTZAN and SMC. The results position CASM as a conservative F1–continuity operating point rather than a universal replacement for DBN. An exploratory calibration-scale study further shows metric-dependent saturation and decreasing sensitivity to fold composition.

Index Terms— beat tracking, downbeat, content-aware adaptation, semi-Markov decoding, post-processing, local periodicity

1. INTRODUCTION

The final decoder in a beat tracker is deceptively consequential. A neural network can assign high framewise probability to genuine beats while still producing duplicates, gaps, metrical-level switches, or downbeats that are inconsistent with its beat grid. Event-level F-measure rewards accurate local placement, whereas continuity measures additionally ask whether one metrical interpretation persists through time. This creates a practical tension: weak decoding respects a capable network but may yield an unstable clock; strong decoding restores a clock but can overwrite correct, expressive, or unusual evidence.

One prevalent structured decoder is the dynamic Bayesian network (DBN) [1]. Its latent tempo, position, and meter states are valuable precisely because they can repair noisy activations. The permitted tempo support, meter set, and transition stiffness, however, are configured globally. The posterior adapts its state path to the input, but the form and strength of the temporal assumptions do not become more cautious when the input itself reveals half/double-tempo ambiguity. Re-optimizing those settings on a target corpus is legitimate development-set model selection, yet a test- or target-informed recipe is no longer evidence of out-of-the-box transfer. Recent post-processing-free trackers show that strong networks can avoid some DBN errors [2, 3], but minimal peak picking can sacrifice sequence continuity. Recent failure analysis on the deliberately difficult SMC corpus makes this trade-off especially visible [4].

We ask whether the temporal prior itself can listen to the current activation. Our answer is *CASM*, a Context-Aware Semi-Markov post-processor. CASM constructs a sparse graph whose nodes are activation maxima and whose edges are complete inter-beat segments. At both endpoints of an edge, it estimates a local period and a best-versus-competing-period margin. Clear periodic evidence creates a narrow, strong duration potential; ambiguous evidence simultaneously weakens and broadens that potential. CASM is consequently an automatic correction layer in a precise, limited sense: after one global calibration, it requires neither track metadata nor per-track decoder adjustment, while its constraints remain input-dependent at inference.

Our contributions are threefold:

- We formulate maximum-score beat decoding on a sparse semi-Markov graph of activation-supported candidates. Each duration potential uses both endpoint contexts and is conditioned on their ambiguity.
- We connect beat decoding to a beat-synchronous variable-meter decoder and two explicit safeguards, yielding a deterministic plug-in that does not retrain the upstream tracker.
- We study one frozen operating point across three different activation backbones and analyse how the amount and composition of decoder-calibration data affect transfer, while separating exploratory selection from clean evaluation claims.

2. RELATED WORK

2.1. From activations to event sequences

Dynamic programming has long combined onset evidence with inter-beat regularity [5]. Korzeniowski *et al.* proposed a Bayesian beat-position extractor around a dominant beat interval [6]; its implementation is exposed by `madmom` under the historical name `CRFBeatDetectionProcessor`, although the paper is not the learned linear-chain CRF of Fillon *et al.* [7]. The learned CRF branch was extended to downbeats [8] and, in 2019, to skip-chain musical structure and Afro-Latin microtiming [9, 10]. Resonating comb filters form a different branch: they estimate a dominant global tempo from recurrent activations [11], after which events are aligned to that periodicity. The 2019 multi-task tempo–beat model [12] extended neural training rather than this comb-filter event decoder, and retained structured inference. DBNs instead enumerate tempo–phase–meter states and decode their most likely trajectory [1, 13]; they remain common in recent high-performing systems.

2.2. Local, segmental, and modern alternatives

Reliability-informed tracking established the principle that temporal commitment should depend on evidence quality [14]. More recently, a compact one-dimensional semi-Markov state space represented time-varying rhythmic jumps for causal joint tracking [15]. Particle filters have long maintained online tempo hypotheses [16], and BeatNet/BeatNet+ extend that branch to joint real-time rhythm analysis [17,18]. PLPDP is CASM's closest conceptual predecessor: it derives predominant local pulse information and uses dense-frame dynamic programming for expressive tempo [19]. Its real-time continuation improves controllability [20], and dPLP makes local-pulse estimation differentiable [21].

CASM does not claim novelty for local periodicity, dynamic programming, or semi-Markov inference separately. Unlike PLPDP's dense frame grid, it scores only segments joining retained activation maxima and conditions each segment on the local period *and ambiguity at both endpoints*. Ambiguity modulates both penalty strength and tolerance, rather than merely supplying a preferred period. Unlike the one-dimensional causal semi-Markov model, CASM is an offline sparse event graph with explicit fallback behaviour. Unlike a DBN, it does not maintain a dense tempo–phase state lattice; unlike a semi-CRF [22], its potentials are not learned or normalized.

Another response to difficult rhythm is to adapt the tracker with user annotations [23–27]. That line is complementary: it deliberately imports target-specific supervision. CASM instead recomputes its constraint from unlabelled test activations and leaves upstream weights fixed. At the other extreme, recent work reformulates the prediction task itself [28]; masked diffusion learns coherent sequences through iterative refinement [29], while BeatFCOS predicts intervals as objects with Soft-NMS [30]. Knowledge-driven metrical hypotheses have also been introduced during self-supervised pre-training [31]. These are modern alternatives to traditional post-processing, but require a changed or retrained predictor; CASM targets the still-useful deployment setting in which only a frozen activation interface is available.

3. METHODOLOGY

3.1. CASM: Constraint-Adaptive Semi-Markov Decoding

Let a frozen upstream tracker produce beat and downbeat logits z_t^{b} and z_t^{d} at frame rate F. We propose *CASM*, a deterministic segmental MAP decoder that converts these framewise scores into a musically plausible event sequence. Its coefficients are fixed globally, but its duration constraint is recomputed from the input at each candidate event. Thus, CASM requires no track-specific BPM, meter, or regularization setting: the activation itself determines both the locally expected inter-beat interval and how strongly that expectation should be enforced.

3.1.1. *Relation to structured beat decoders.*

CASM inherits the classical separation between an acoustic activation model and a structured temporal decoder. Earlier systems used a whole-track beat interval in a Bayesian beat-position model—implemented as the "CRF" processor in `madmom`— [6], learned dense period–position labels with a linear-chain CRF [7], or enumerated latent tempo and bar phase in a DBN [1]. CRFs were

Fig. 1. Workflow for beat post-processing...

subsequently extended to downbeats, long-range section consistency, and genre-specific microtiming [8–10]. Resonating comb filters instead estimate a dominant global periodicity and do not themselves select a beat-event path [11].

CASM is not a new name for any of these models. It has neither a dense tempo–phase state grid nor a learned, normalized CRF distribution; its only latent object is a sparse path through activation-supported events. It also differs from the causal one-dimensional semi-Markov pointer of [15], which maintains a framewise posterior and adapts a jump-back distribution online. The closest conceptual predecessor is PLPDP [19]: both derive time-varying periodicity from an activation before dynamic programming. PLPDP, however, decodes on a dense frame grid using a current-frame PLP confidence. CASM (i) never creates a beat away from a retained activation maximum, (ii) conditions each segment on both endpoints, (iii) uses ambiguity to change both penalty strength and tolerance, and (iv) adds beat-count and downbeat-agreement safeguards. Following the principle of reliability-informed tracking [14], CASM lets uncertain evidence weaken structural commitment; unlike that work's track-level predicted accuracy, c_i is a deterministic local period margin. Motivated also by the octave and continuity failures exposed on SMC [4], the single-path CASM does not claim to solve multi-hypothesis tempo inference; instead, it detects a locally ambiguous period and then deliberately backs off from enforcing it.

3.1.2. *Candidate-event graph.*

Let $a_t = \mathrm{sigmoid}(z_t^{\mathrm{b}})$. We retain its one-dimensional local maxima (including valid boundary maxima) whose $a_t \geq \theta_{\mathrm{c}}$, producing an ordered candidate set $\mathcal{C} = \{t_1, \ldots, t_N\}$. An edge (i, j) represents the entire candidate-to-candidate segment and is admissible only if $\Delta_{ij} = (t_j - t_i)/F \in [60/B_{\max}, 60/B_{\min}]$. For a path $\pi = (i_1, \ldots, i_K)$, CASM maximizes

$$\mathcal{S}(\pi) = \sum_{k=1}^{K} e_{i_k} - \sum_{k=2}^{K} D_{i_{k-1}, i_k}, \tag{1}$$

where e_i is activation evidence and D_{ij} is an explicit segment-duration potential. This variable-duration event formulation is the precise sense in which CASM is semi-Markov [22]; it is not a generative hidden semi-Markov model.

3.1.3. *Local period and ambiguity.*

For each candidate i, CASM evaluates integer lags $\ell \in \mathcal{L} = \{\ell_{\min}, \ldots, \ell_{\max}\}$ in a centred local window $\mathcal{W}_i$. Its normalized, comb-like autocorrelation score is

$$q_i(\ell) = \frac{\langle a_u a_{u-\ell} \rangle_{u \in \mathcal{W}_i}}{\sqrt{\langle a_u^2 \rangle_{u \in \mathcal{W}_i} \langle a_{u-\ell}^2 \rangle_{u \in \mathcal{W}_i} + \epsilon_q}} \left(\frac{\ell_{\min}}{\ell} \right)^{\beta}, \tag{2}$$

where values outside the signal are zero and the last factor is a mild fast-tempo prior. We select $\ell_i^\star = \arg\max_{\ell \in \mathcal{L}} q_i(\ell)$ and $\tau_i = \ell_i^\star / F$. To expose half/double-time and neighbouring periodicities, the competing score excludes a ± 2-lag neighbourhood:

$$c_i = \operatorname{clip}_{[0,1]} \left(\frac{q_i(\ell_i^\star) - \max_{\ell:\, |\ell - \ell_i^\star| > 2} q_i(\ell)}{|q_i(\ell_i^\star)| + \epsilon_c} \right). \tag{3}$$

The margin c_i measures local periodicity separation, not calibrated correctness. Both τ_i and c_i are median-filtered over five consecutive candidates.

3.1.4. *Constraint-adaptive beat decoding.*

For edge (i, j), define endpoint-symmetric context $\bar{\tau}_{ij} = \sqrt{\tau_i \tau_j}$ and $c_{ij} = \sqrt{c_i c_j}$. The duration cost is

$$D_{ij} = \frac{\lambda c_{ij}}{2} \left[\frac{\log(\Delta_{ij} / \bar{\tau}_{ij})}{\sigma_0 + (1 - c_{ij}) \sigma_{\mathrm{u}}} \right]^2. \tag{4}$$

A well-separated local period therefore produces a strong, narrow temporal constraint. When several metrical levels are similarly plausible, the same margin both lowers the multiplier and widens the allowed deviation. This is the method's adaptation mechanism: the decoder can remove activation-supported duplicates or select a weaker retained peak without imposing a dubious grid.

With $e_j = \operatorname{clip}(z_{t_j}^{\mathrm{b}} / T, -L, L)$, the offline recurrence for Eq. (1) is

$$V_j = e_j + \max \left\{ 0, \max_{i < j:\, (i,j) \text{ admissible}} (V_i - D_{ij}) \right\}. \tag{5}$$

Backtracking begins at $\arg\max_j V_j$; the zero branch permits the best path to start after an unfavourable prefix. We also form the direct path of [3] by seven-frame max pooling, a zero-logit threshold, and one-frame peak deduplication. If the structured-to-direct event-count ratio falls outside $[r_{\min}, r_{\max}]$, CASM returns the direct path. This deterministic sanity check is not a learned router.

3.1.5. *Beat-synchronous downbeat decoding.*

Beat and downbeat inference are sequential. Given the decoded beats $\widehat{\mathcal{B}} = \{\widehat{b}_n\}$, we only score downbeats on this grid. For a candidate bar length $m \in \mathcal{M}$, let $g_n = \operatorname{clip}(z_{\widehat{b}_n}^{\mathrm{d}} / T_{\mathrm{d}}, -L, L) + \rho$. The meter recurrence is

$$H_{n,m} = g_n + \max_{m' \in \mathcal{M}} \left[H_{n-m, m'} - \eta \, \mathbf{1}(m' \neq m) \right], \tag{6}$$

with initial phases in the first $\max(\mathcal{M})$ beats and termination in the last $\max(\mathcal{M})$ beats. It can therefore change bar length while penalizing gratuitous meter switches. Independently, direct downbeat maxima are snapped to $\widehat{\mathcal{B}}$. We retain the meter path only when its one-to-one event F-measure against this snapped path, at 70-ms tolerance, is at least γ; otherwise the snapped path is returned. All output downbeats thus remain on the decoded beat grid.

3.1.6. *Frozen operating point and supervision boundary.*

CASM-Frozen-4F uses $F = 50$ Hz, $\theta_{\mathrm{c}} = 0.03$, $(B_{\min}, B_{\max}) = (30, 300)$ BPM, an 8-s local window, $\beta = 0.15$, $(T, L) = (2, 6)$, $(\lambda, \sigma_0, \sigma_{\mathrm{u}}) = (4, 0.12, 0.40)$, and $[r_{\min}, r_{\max}] = [0.85, 1.8]$; $\epsilon_q = 10^{-8}$ and $\epsilon_c = 10^{-6}$. Downbeat inference uses $\mathcal{M} = \{2, 3, 4, 5, 6, 7\}$, $(T_{\mathrm{d}}, \eta, \rho) = (4, 3, 0)$, and $\gamma = 0.6$. These scalar hyperparameters were selected on development folds $\{2, 7, 1, 3\}$ and then frozen for every track and every reported upstream tracker. The choice to foreground this four-fold operating point after the fold-scaling analysis is post hoc; the separately locked seven-fold comparator is reported as such in the experimental protocol. Hence CASM is input-adaptive and free of per-track manual tuning, but not parameter-free. It has no learned weights and receives neither annotated beat times nor inter-beat intervals at inference.

4. EXPERIMENTS

4.1. Data, backbones, and metrics

We evaluate on GTZAN ($N = 993$) using the annotations and exclusions of Beat This [3], and on the deliberately difficult SMC corpus ($N = 217$) [32]; SMC provides beat but not downbeat annotations. We use three 50-Hz activation backbones: the Beat This spectrogram Transformer [3], the lightweight multi-scale convolutional backbone MSCNN-lite [33], and the TCN of [34]. The decoder receives only their framewise beat and downbeat logits.

For GTZAN, `final0`, `final1`, and `final2` are three independently trained full-data backbone seeds. Their training excludes GTZAN but includes SMC. We first macro-average over the same 993 tracks and then average the three seed-level scores. For SMC, using those full-data checkpoints would leak training examples; we instead use the corresponding held-out checkpoint for each of eight folds. Seven folds contain 27 tracks and the eighth contains 28; we average their fold-level macro scores. CASM itself is deterministic and has neither a neural checkpoint nor a random seed.

Following the established protocol, the first 5 s are trimmed and `mir_eval` computes F1 with ± 70 ms tolerance, CMLt, and AMLt [35]. CMLt requires a continuously correct metrical level; AMLt additionally accepts the standard off-beat and half/double-level interpretations. Seed variation on GTZAN and fold variation on SMC have different replication units and are descriptive, not confidence intervals.

4.2. Decoder baselines

Direct is the minimal peak picker defined in Sec. 3. *DBN* is the conventional joint `madmom` configuration: meters $\{3, 4\}$, 55–215 BPM, transition weight 100, observation weight 16, and threshold 0.05 [1]. *DBN-SMC-opt* is deliberately diagnostic. Its sequential beat and bar stages were selected from 753 configurations on the TCN SMC 8-fold OOF activations, maximizing beat F1 with CMLt/AMLt tie-breaks, and only then frozen for the three backbones and GTZAN. It uses 50–215 BPM, 30 tempo states, transition/observation weights 7/20, threshold 0.3, meters $\{3, 4\}$, bar-observation weight 100, and meter-switch probability 10^{-7}. It is neither a clean SMC test nor a published SMC-specific recipe from [34].

The *CRF* row uses `madmom`'s historically named `CRFBeatDetectionProcessor` [6] at 30–300 BPM with interval width 0.18; it should not be confused with the learned CRF in [7]. *CombFilter* uses `madmom`'s global tempo/iterative-alignment processor [11]. CRF and CombFilter decode downbeat activations at 8–100 BPM and snap their outputs to their respective beat grids.

4.3. Frozen operating point and scale analysis

Every CASM row in Sec. 5 uses exactly one explicitly loaded Frozen-4F configuration:

$$
\begin{aligned}
&F = 50, \quad \theta_{\mathrm{c}} = 0.03, \quad (B_{\min}, B_{\max}) = (30, 300),\\
&|\mathcal{W}| = 8\,\mathrm{s}, \quad \beta = 0.15, \quad (T, L) = (2, 6),\\
&(\lambda, \sigma_0, \sigma_{\mathrm{u}}) = (4, 0.12, 0.40), \quad [r_{\min}, r_{\max}] = [0.85, 1.8],\\
&\mathcal{M} = \{2, \ldots, 7\}, \quad (T_{\mathrm{d}}, \eta, \rho, \gamma) = (4, 3, 0, 0.6).
\end{aligned}
$$

Its scalar settings were selected on the union of SMC folds $\{2, 7, 1, 3\}$ and then reused for all tracks and all backbones. Thus context-aware means input-conditioned inference, not test-time parameter updates: CASM has validation-selected parameters, but no annotated beat, inter-beat-interval, BPM, or meter input at inference.

The choice to foreground 4F was made *after* inspecting the calibration sensitivity analysis and is therefore exploratory. The seven-fold configuration is the only comparator locked before GTZAN inspection; it differs from 4F only in $\sigma\_0 = 0.15$. For the scale study, SMC fold 0 ($N = 27$) is never used for selection. From folds 1–7 we enumerate every 1-, 2-, and 4-fold calibration subset (7, 21, and 35 combinations) plus the seven-fold union, and evaluate all locked configurations on the same SMC fold 0 and fixed GTZAN/`final1` panel. The latter panel is test-conditioned and is used only to describe sensitivity, not to claim a held-out optimum. [1]

5. RESULTS

5.1. Main Result

As presented in Table **??**, our proposed multi-task model achieves SOTA performance in dynamics and change point estimation, while performing competitively on the remaining tasks. The effectiveness of the multi-task learning paradigm is underscored by the model's superior performance relative to its single-task counterpart using the same BSSL features. This includes significant F1 score improvements in dynamics (+3.8%), change points (+5.1%), beats (+0.1%),

[1]The DBN setup "SMC-opt" is optimized for the best average scores in SMC dataset, which modify (B_min, B_max) from $[55, 215]$ by default to $[30, 215]$. If tune more parameters per song can achieve higher scores as reported in [4]. However, this will broken our test proctorol, so we adpot global setup instead.

Table 1. Comparison of beat-tracking post-processors across different backbone models on GTZAN and SMC. DBN post-processor's default $bpm = [55, 215]$ was optimized for general music. SMC-opt $bpm = [30, 300]$ is optimized to fit SMC data, investigated by [4].

Method	GTZAN (Beat / Downbeat)						SMC (Beat)		
	F1	CMLt	AMLt	F1	CMLt	AMLt	F1	CMLt	AMLt
BeatThis (direct) [3]	89.1	79.8	89.8	78.3	67.2	79.1	62.7	51.4	61.0
w. CASM	**+0.2**	+0.5	+0.6	**+0.5**	+3.0	+5.9	+0.2	+2.4	+3.3
w. CASM, SMC-opt	+0.1	+0.3	+0.4	**+0.5**	+2.9	+5.9	**+0.3**	**+2.5**	+3.2
w. DBN [1]	−1.0	+0.7	+1.3	−0.9	**+6.2**	**+8.7**	−5.2	−3.9	+1.5
w. DBN, SMC-opt	−0.5	−0.9	+0.2	−1.5	−0.6	+1.1	−1.6	+0.4	**+3.5**
w. PLPDP [19]	−0.1	**+0.8**	+0.1	−0.2	+0.1	−0.1	−1.4	−4.7	+0.5
w. PLPDP, SMC-opt	+0.1	+0.3	+0.3	+0.1	+0.2	+0.2	−0.9	−3.5	−0.3
w. CombFilter [11]	−4.0	−3.6	+0.1	−3.3	+3.2	+7.6	−7.1	+1.1	+3.1
w. CRF [6]	−0.6	−0.3	**+1.4**	−1.4	+1.9	+5.5	−1.2	+1.4	+1.2
MSCNN (direct) [33]	86.1	73.4	79.9	70.8	53.1	63.1	57.2	42.0	48.4
w. CASM	**+0.4**	+1.5	+2.4	+1.7	+6.8	+11.7	**+0.5**	+3.3	+4.4
w. DBN	−0.6	**+4.0**	**+7.2**	**+1.8**	**+15.8**	**+21.2**	−4.9	+0.3	+3.9
w. DBN, SMC-opt	−1.1	+0.2	+5.9	−0.9	+9.5	+14.6	+0.3	**+3.5**	**+6.8**
TCN (direct) [34]	87.1	73.8	81.7	62.4	24.6	59.9	52.9	30.1	36.0
w. CASM	**+0.3**	**+1.3**	+1.6	+2.3	+19.4	+14.0	+0.3	+3.8	+5.2
w. DBN	−2.6	−0.7	**+5.0**	**+4.9**	**+23.2**	**+22.9**	−0.9	−7.0	+2.3
w. DBN, SMC-opt [4]	−1.9	−0.8	+4.4	−0.2	+8.9	+9.3	**+1.7**	**+4.5**	**+8.2**

and downbeats (+10.2%). Beyond these quantitative improvements, the multi-task model offers considerable practical utility. It operates within a highly parameter-efficient framework (0.5 M vs. 4× single-task 0.4 M) and can utilize its own predicted beat positions for post-processing, enabling practical application on unannotated audio.

Our analysis also reveals a strong task-feature dependency: BSSL features are optimal for dynamics, change points, and beats estimation, whereas log-Mel features are preferable for downbeat tracking. A primary advantage of BSSL is its compactness (22 Bark bins vs. 128 Mel bins in our STFT setup). Within our multi-task multi-scale network, which relies on convolutional residual blocks, this smaller input dimension can reduce the model's trainable parameters from 14.7 M to just 0.5 M. This significantly smaller footprint enables the model to process longer audio sequences, directly benefiting tasks that require long-term temporal information and highlighting BSSL's potential for a wider range of musical applications with long-term dependencies.

5.2. What calibration-scale sensitivity shows

The fixed-panel sensitivity analysis does not show that 4F is optimal, nor that more data hurts machine learning. CASM learns no network weights here: the independent variable is the amount of data used to select global decoder scalars. On the held-out SMC panel, the family centres of F1, CMLt, and AMLt rise from 1F to 7F. On GTZAN, beat F1 and downbeat F1/CMLt peak at 2F, while beat CMLt/AMLt and downbeat AMLt continue to 7F. Hence the defensible conclusion is metric-dependent diminishing returns rather than a universal best scale.

There is nevertheless a useful stability result. From 1F to 2F to 4F, the across-combination population spread decreases in all nine panels: more calibration data makes the selected decoder less sensitive to which SMC folds are present, even when the external point estimate does not improve monotonically. We retain 4F as a balanced, post-hoc operating point for the present analysis. Its near-identical seven-fold comparator is the appropriate pre-GTZAN locked reference; a new external test set or nested selection is needed to establish superiority.

Table 2. Comparison with recent state-of-the-art beat and downbeat tracking systems on GTZAN (mean over 3 seeds) and SMC dataset (mean over 8-fold). DBN uses the default setup, which is already optimized for general music.

Method	GTZAN (Beat / Downbeat)						SMC (Beat)		
	F1	CMLt	AMLt	F1	CMLt	AMLt	F1	CMLt	AMLt
HN-MERT w. DBN [36]	**89.7**	81.4	**94.3**	77.4	71.6	87.2	61.7	–	–
HN-MusicFM w. DBN	89.2	80.9	93.7	**79.8**	73.2	**89.5**	60.6	–	–
BeatFM-MERT w. DBN [37]	<u>89.5</u>	<u>82.7</u>	<u>93.9</u>	76.7	70.2	85.4	61.3	–	–
BeatFM-MusicFM w. DBN	89.1	80.6	93.5	<u>79.6</u>	<u>74.4</u>	<u>88.7</u>	60.5	–	–
BeatMamba w. DBN [38]	88.9	82.3	**94.3**	–	–	–	58.7	46.8	62.4
BeatKAN w. DBN [39]	88.2	78.1	92.3	–	–	–	59.8	48.4	<u>64.0</u>
MDM-BeatThis (direct) [29]	**89.7**	**82.9**	92.5	79.5	**76.4**	88.5	61.0	48.9	**64.3**
BeatThis (direct) [3]	89.1	79.8	89.8	78.3	67.2	79.1	<u>62.7</u>	<u>51.4</u>	61.0
BeatThis w. DBN	88.1	80.5	91.1	77.4	73.4	87.8	57.5	47.5	62.5
BeatThis w. CASM, proposed	89.3	80.2	90.4	78.9	70.4	85.1	**63.0**	**53.8**	**64.3**

Following the evaluation protocols of prior work [3, 36, 37], we report the mean over three random seeds for GTZAN and the median over eight folds for SMC. For compactness, the table presents only these aggregated scores; complete seed- and fold-level results are provided in our GitHub repository. For CASM, the backbone predictions are generated in an eight-fold OOF setting, while the decoder configuration is selected using folds $\{2, 7, 1, 3\}$; therefore, this row should not be interpreted as strict nested decoder-selection OOF. CASM uses the Frozen-4F configuration selected using folds $\{2, 7, 1, 3\}$.

5.3. Limitations and decisive next tests

Three limitations bound the current claim. First, SMC activations are backbone-level OOF, but the single 4F decoder is not selected in a nested outer loop. Second, CASM searches 30–300 BPM while the default DBN searches 55–215 BPM. Recent SMC analysis reports many references below that lower limit [4]; a matched-support DBN is therefore required before attributing an SMC difference solely to CASM's ambiguity mechanism. Third, the present aggregate tables do not isolate the sparse graph, strength-versus-width ambiguity gate, or the two safeguards. A direct PLPDP baseline, fixed-period DP, component ablations, and per-corpus fallback and meter-path acceptance rates are the decisive experiments. Track-paired bootstrap or randomization tests should accompany them; three training seeds or eight folds alone do not justify significance claims.

6. CONCLUSION AND FUTURE WORK

CASM revisits structured beat decoding as an inductive bias conditioned on each input, not as a return to a fixed global grid. It joins activation-supported events with locally centred segment potentials and deliberately relaxes those potentials when periodic evidence is ambiguous. Once globally calibrated, the same deterministic module runs without upstream retraining, user annotation, or per-track tempo and meter settings. Current evidence supports a conservative F1–continuity alternative to Direct and DBN decoding, not universal dominance. The next version should use strict nested or leave-one-dataset-out calibration, matched tempo-support baselines, PLPDP and mechanism ablations, and explicit safeguard statistics. Multi-hypothesis period paths, causal decoding, and a differentiable duration layer are natural extensions, as is evaluation on every recent tracker that releases compatible activations.

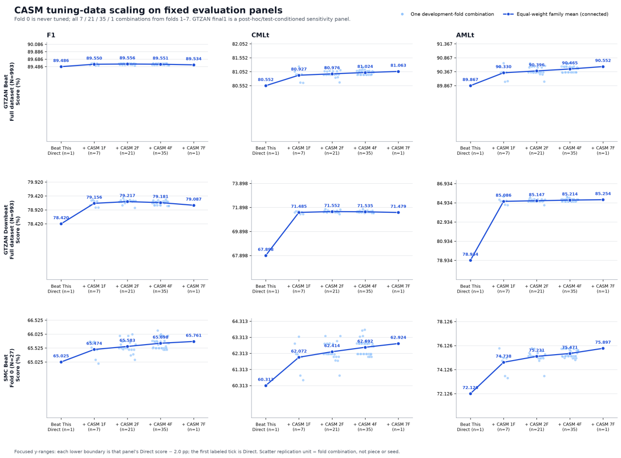

Fig. 2. Here is explain how to train Semi-Markov... about the data scaling problem

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